

The functors $\mathrm{Ext}_Z^\bullet(X; F, G)$ and $\underline{\mathrm{Ext}}_Z^\bullet(F, G)$
SGA 2, Exposé VI

English draft

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About this draft

This is an unofficial English translation of Exposé VI, “The functors $\mathrm{Ext}_{\mathbb{Z}}^{\bullet}(X; F, G)$ and $\underline{\mathrm{Ext}}_{\mathbb{Z}}^{\bullet}(F, G)$ ”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé VI. The functors $\mathrm{Ext}_Z^\bullet(X; F, G)$ and $\underline{\mathrm{Ext}}_Z^\bullet(F, G)$	1
1. Generalities	1
2. Applications to quasi-coherent sheaves on preschemes	2
Bibliography	3

EXPOSÉ VI

The functors $\text{Ext}_Z^\bullet(X; F, G)$ and $\underline{\text{Ext}}_Z^\bullet(F, G)$

1. Generalities

1.1. Let $(X, \mathcal{O}_X)$ be a ringed space and Z a locally closed subset of X . Let F and G be $\mathcal{O}_X$ -Modules; one denotes by $\text{Ext}_Z^i(X; F, G)$ (resp. $\underline{\text{Ext}}_Z^i(F, G)$) the i -th derived functor of the functor $G \mapsto \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G))$ (resp. $\Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G))$).

LEMMA 1.2. *The sheaf $\underline{\text{Ext}}_Z^i(F, G)$ is canonically isomorphic to the sheaf associated to the presheaf $U \mapsto \text{Ext}_{Z \cap U}^i(U; F|_U, G|_U)$.*

This follows from ([?], 3.7.2) and from the fact that $\Gamma(U; \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G)))$ is canonically isomorphic to $\Gamma_{Z \cap U}(\underline{\text{Hom}}_{\mathcal{O}_{X|U}}(F|_U, G|_U))$.

THEOREM 1.3 (Excision theorem). *Let V be an open subset of X containing Z ; one then has an isomorphism of cohomological functors*

$$(1.3.1) \quad \text{Ext}_X^\bullet(X; F, G) \simeq \text{Ext}_V^\bullet(V; F|_V, G|_V)$$

Indeed, if $G^\bullet$ is an injective resolution of G , then $G|_V^\bullet$ is an injective resolution of $G|_V$. The theorem follows immediately.

1.4. Let $\mathcal{O}_{X,Z}$ be the $\mathcal{O}_X$ -Module defined by the following conditions ([?], 2.9.2): $\mathcal{O}_{X,Z}|_{X-Z} = 0$ and $\mathcal{O}_{X,Z}|_Z = \mathcal{O}_{X|Z}$. One has seen that for every $\mathcal{O}_X$ -Module H , there exists a functorial isomorphism: $\Gamma_Z(H) \simeq \text{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, H)$. One therefore deduces functorial isomorphisms in F and G :

$$(1.4.1) \quad \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G)) \simeq \text{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, \underline{\text{Hom}}_{\mathcal{O}_X}(F, G)),$$

$$(1.4.2) \quad \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G)) \simeq \text{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G),$$

$$(1.4.3) \quad \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G)) \simeq \text{Hom}_{\mathcal{O}_X}(F, \underline{\text{Hom}}_{\mathcal{O}_X}(\mathcal{O}_{X,Z}, G)) = \text{Hom}_{\mathcal{O}_X}(F, \Gamma_Z(G)).$$

It follows in particular from (1.4.2) that there exists an isomorphism ∂ -functorial in F and G

$$\theta: \text{Ext}_{\mathcal{O}_X}^i(\mathcal{O}_{X,Z} \otimes_{\mathcal{O}_X} F, G) \xrightarrow{\sim} \text{Ext}_Z^i(X; F, G).$$

1.5. By definition the functor $G \mapsto \Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G))$ is the composite of the functor $G \mapsto \underline{\text{Hom}}_{\mathcal{O}_X}(F, G)$ and of the functor Γ_Z . As Γ_Z is left exact (I I.1.9) and as $\underline{\text{Hom}}_{\mathcal{O}_X}(F, G)$ is flasque if G is injective, and as Γ_Z is exact on flasques (I I.2.12), it follows from ([?], 2.4.1) that there exists a spectral functor abutting to $\text{Ext}_Z^\bullet(X; F, G)$ whose initial term is $H_Z^p(X, \underline{\text{Ext}}_{\mathcal{O}_X}^q(F, G))$.

On the other hand, it follows from (1.4.3) that $\Gamma_Z(\underline{\text{Hom}}_{\mathcal{O}_X}(F, G))$ is the composite of Γ_Z and of the functor $H \mapsto \text{Hom}_{\mathcal{O}_X}(F, H)$.

Since the functor Γ_Z transforms injectives into injectives (I I.1.4), it follows from ([?], 2.4.1) that there exists a spectral functor abutting to $\text{Ext}_Z^\bullet(X; F, G)$ whose initial term is $\text{Ext}_{\mathcal{O}_X}^p(X; F, \underline{H}_Z^q(G))$.

Finally, it follows from (1.4.2) and from the spectral sequence of the Ext , that there exists a spectral functor abutting to $\text{Ext}_Z^\bullet(X; F, G)$ whose initial term is $H^p(X; \underline{\text{Ext}}_Z^\bullet(F, G))$. Hence the

THEOREM 1.6. *There exist three spectral functors abutting to $\text{Ext}_Z^\bullet(X; F, G)$ whose initial terms are respectively*

$$(1.6.1) \quad H_Z^p(X, \underline{\text{Ext}}_{\mathcal{O}_X}^q(F, G))$$

$$(1.6.2) \quad H^p(X, \underline{\text{Ext}}_Z^q(F, G))$$

$$(1.6.3) \quad \text{Ext}_{\mathcal{O}_X}^p(X; F, \underline{H}_Z^q(G)).$$

1.7. Now let Z' be a closed subset of Z and let $Z'' = Z - Z'$. One has an exact sequence

$$(1.7.1) \quad 0 \longrightarrow \mathcal{O}_{X, Z''} \longrightarrow \mathcal{O}_{X, Z} \longrightarrow \mathcal{O}_{X, Z'} \longrightarrow 0$$

which generalizes the exact sequence of ([?], 2.9.3). This exact sequence splits locally; one therefore has, for every $\mathcal{O}_X$ -Module F , another exact sequence:

$$(1.7.2) \quad 0 \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X, Z''} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X, Z} \longrightarrow F \otimes_{\mathcal{O}_X} \mathcal{O}_{X, Z'} \longrightarrow 0$$

Now let G be an $\mathcal{O}_X$ -Module; if one applies the functor $\text{Hom}_{\mathcal{O}_X}(\bullet, G)$ to the exact sequence (1.7.2), one deduces from (1.4.2) and from the exact sequence of the Ext the following theorem:

THEOREM 1.8. *Let Z be a locally closed subset of X , Z' a closed subset of Z and $Z'' = Z - Z'$. One then has an exact sequence functorial in F and G :*

$$\begin{aligned} 0 \longrightarrow \text{Hom}_{Z'}(F, G) \longrightarrow \text{Hom}_Z(F, G) \longrightarrow \text{Hom}_{Z''}(F, G) \longrightarrow \text{Ext}_{Z'}^1(F, G) \longrightarrow \cdots \\ \cdots \text{Ext}_Z^i(F, G) \longrightarrow \text{Ext}_{Z''}^i(F, G) \longrightarrow \text{Ext}_{Z'}^{i+1}(F, G) \longrightarrow \cdots \end{aligned}$$

COROLLARY 1.9. *Let Y be a closed subset of X and let $U = X - Y$. One then has an exact sequence functorial in F and G :*

$$\begin{aligned} 0 \longrightarrow \text{Hom}_Y(F, G) \longrightarrow \text{Hom}_{\mathcal{O}_X}(F, G) \longrightarrow \text{Hom}_{\mathcal{O}_{X|U}}(F|_U, G|_U) \longrightarrow \text{Ext}_Y^1(F, G) \longrightarrow \cdots \\ \cdots \text{Ext}_{\mathcal{O}_X}^i(F, G) \longrightarrow \text{Ext}_{\mathcal{O}_{X|U}}^i(F|_U, G|_U) \longrightarrow \text{Ext}_Y^{i+1}(F, G) \longrightarrow \cdots \end{aligned}$$

This corollary is an immediate consequence of theorem (1.3) and of theorem (1.8).

2. Applications to quasi-coherent sheaves on preschemes

PROPOSITION 2.1. *Let X be a locally noetherian prescheme; for every locally closed subset Z of X , for every coherent Module F and every quasi-coherent Module G on X , the $\underline{\text{Ext}}_Z^i(F, G)$ are quasi-coherent.*

One shows, as for (1.6.3), that the Modules $\underline{\text{Ext}}_Z^i(F, G)$ are the abutment of a spectral sequence with initial term $\underline{\text{Ext}}_{\mathcal{O}_X}^p(F, \underline{H}_Z^q(G))$. By (II, cor. II.3), the $\underline{H}_Z^q(G)$ are quasi-coherent, and hence so are the $\underline{\text{Ext}}_{\mathcal{O}_X}^p(F, \underline{H}_Z^q(G))$, since F is coherent. The proposition then follows immediately.

2.2. Now let Y be a closed subscheme of X and $\mathcal{I}$ an ideal of definition of Y . Let m and n be integers such that $m \geq n \geq 0$; one denotes by $i_{n,m}$ the canonical map $\mathcal{O}_{Y_m} = \mathcal{O}_X/\mathcal{I}^{m+1} \rightarrow \mathcal{O}_X/\mathcal{I}^{n+1} = \mathcal{O}_{Y_n}$ and by j_n the map $\mathcal{O}_{X,Y} \rightarrow \mathcal{O}_{Y_n}$. The $(\mathcal{O}_{Y_n}, i_{n,m})$ form a projective system and the j_n are compatible with the $i_{n,m}$.

Applying the functor $\text{Ext}_{\mathcal{O}_X}^i(F \otimes \bullet, G)$, one deduces from this a morphism

$$\varphi': \varinjlim_n \text{Ext}_{\mathcal{O}_X}^i(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \text{Ext}_{\mathcal{O}_X}^i(X; F \otimes \mathcal{O}_{X,Y}, G);$$

it is a morphism of cohomological functors in G . The morphism

$$\varphi: \varinjlim_n \text{Ext}_{\mathcal{O}_X}^i(X; F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \text{Ext}_Y^i(X; F, G)$$

composite of φ' and of θ (cf. 1.4) is therefore itself also a functor of cohomological morphisms in G .

One defines similarly

$$\underline{\varphi}: \varinjlim_n \underline{\text{Ext}}_{\mathcal{O}_X}^i(F \otimes \mathcal{O}_{Y_n}, G) \longrightarrow \underline{\text{Ext}}_Y^i(X; F, G)$$

THEOREM 2.3. *Let X be a locally noetherian prescheme, Y a closed subset of X defined by a coherent ideal $\mathcal{I}$, F a coherent Module, G a quasi-coherent Module. Then,*

- a) φ is an isomorphism.
- b) *If X is noetherian, φ is an isomorphism.*

The proof of b) being almost word for word that of (II II.6b)), thanks to the spectral sequence 1.6.2, we shall not reproduce it.

For the proof of a), one may, by (2.1), assume X affine with ring A , F (resp. G) defined by an A -module M (resp. N) and $\mathcal{I}$ by an ideal I . It suffices to prove that the homomorphism

$$(2.3.1) \quad \varinjlim_n \text{Ext}_A^i(M/I^n M, N) \longrightarrow \text{Ext}_Y^i(X, F, G)$$

deduced from φ is an isomorphism.

Indeed, one may, for $i = 0$, canonically identify the two sides of (2.3.1) with the submodule of $\text{Hom}_A(M, N)$ defined by the elements of $\text{Hom}_A(M, N)$ annihilated by a power of I . One then sees that the homomorphism (2.3.1) is none other than the identity map.

The functor $N \mapsto \varinjlim_n \text{Ext}_A^\bullet(M/I^n M, N)$ is a universal ∂ -functor. We shall show that the same is true of the functor $N \mapsto \text{Ext}_Y^\bullet(M, N)$. Indeed if N is an injective module, by (II.9 and II.11), $\underline{H}_Y^q(N) = 0$ if $q \neq 0$; and by (IV.IV.2.2), $H_Y^0(N)$ is injective.

It then follows that $\text{Ext}_{\mathcal{O}_X}^p(X; M, \underline{H}_Y^q(N)) = 0$ if $p + q \neq 0$; one therefore has, by (1.6.3), $\text{Ext}_Y^i(M, N) = 0$ for $i \neq 0$ and N injective. This completes the proof.

Bibliography

Same references as those listed at the end of Exp. I, cited respectively [?] and [?].